



Research papers

Context matters: A baseline analysis of contextual realities for two community-based monitoring initiatives of water and environment in Europe and Africa

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ABSTRACT

Much attention is being paid to the design and implementation of community-based monitoring initiatives of water and environment, and how to attract more citizens to participate in such initiatives. Although mass participation in collecting and sharing water-related or environmental data and good project design and implementation are key factors for the success of community-based monitoring, these initiatives do not operate in a void. Community-based monitoring initiatives are embedded in, and can influence, existing social, institutional, political and technological settings. In spite of the fact that these contextual factors are not static and constantly change over time, capturing their status quo at the initiation of a community-based monitoring activity delivers critical insights for establishing a sustainable initiative and can be used as a benchmark for assessing its subsequent outcomes and impacts. Yet, the salience of understanding the initial contextual settings is often underestimated, or these are only considered once an initiative has already been established. The objective of this paper is to demonstrate the importance of early stage research about such contextual realities. In order to do so, we employed the CPI Framework (Gharesifard et al., 2019) for conducting a systematic analysis of the baseline situation of two newly established community-based monitoring initiatives, one in the Netherlands and one in Kenya. The case study in the Netherlands focuses on the issue of pluvial flooding in an urban setting, while the Kenyan case study deals with balancing sustainable livelihoods and biodiversity management in a rural area. Our baseline analysis showed that aside from the fact that these initiatives have different thematic foci, there are distinct differences between the two in terms of access to technology, availability and accessibility of data, the institutional arrangements for public participation in decision making processes, and the level of citizen trust in the authorities in charge of managing the respective water-related and environmental issues. Based on the findings of this research, a number of recommendations are provided that can be beneficial for the future development and functioning of the two initiatives.

1. Introduction

Complexity, uncertainty and multi-dimensionality are among the inherent characteristics of environmental problems (Polasky et al., 2011; Reed, 2008) and despite great scientific and technological advancements, solving these challenges has proven to require more than just conventional 'science-centered' ideology (Irwin, 1995). Over the past two decades, the promotion of non-structural measures along with widespread replacement of water and environmental management concepts with governance ideology (Wehn et al., 2015) has highlighted the role of citizens and envisioned their participation as one of the most important stakeholders in the management of natural resources. This

has resulted in paying special attention to citizen science projects and community-based monitoring (hereafter CBM) initiatives of water and environmental resources as innovative approaches that can help address the complex water and environmental challenges of the 21st century. CBM initiatives, on the one hand, have the potential to help enhance the spatial and temporal resolution of shared environmental data, inform environmental decision making, and subsequently help reform water-related and environmental policies; and, on the other hand, they can facilitate the communication between different stakeholders and the co-creation of knowledge.

Much emphasis is being placed on the design and implementation of CBM and how to attract more citizens to participate in such initiatives.

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Although mass participation in collecting and sharing water and environmental data and good project design and implementation are key factors for the success of a CBM initiative, these initiatives do not operate in a void. During the initiation of a CBM initiative, there are already existing institutions, policies, and technological settings in which the initiative will be embedded, with which it will interact, or which it aims to influence. As an example, lessons learned from the WeSenseIt project, an EU-FP7 project for developing CBM initiatives of water and flooding, confirm that such circumstantial factors influenced the participation dynamics and the 'shape and size' of the CBM initiatives in this project (Joshi & Wehn, 2017; Wehn et al., 2015). Although contextual factors¹ are not static and constantly change over time, capturing their status quo at the initiation of a CBM delivers critical insights for establishing a sustainable initiative and for assessing its evolving outcomes and impacts. Nevertheless, the importance of such a baseline analysis is often underestimated or only considered after initiation of a CBM activity. This has resulted in initiatives that discontinue after a short while, whose results have no actual use in policy and decision making, or they even have contributed to creating new forms of exclusion and inequalities within society. We argue that for a CBM to have impact on policy and decision making processes, it is essential to first gain a thorough understanding of the contextual setting in which it will be embedded and operate, since these aspects can strongly affect the design, implementation, outcomes and impacts of the initiative. The aim of this paper is thus to demonstrate the importance of an early analysis of such contextual realities.

Although the role of contextual factors in shaping the design and implementation of Citizen Science and CBM initiatives have been highlighted in the literature (e.g. Dehnen-Schmutz et al., 2016; Irwin, 1995; Phillips et al., 2019; Wehn et al., 2015), until recently, there was no framework that enables a systematic analysis of the contextual factors that affect the establishment and functioning of these initiatives. Gharesifard et al. (2019) proposed a conceptual framework for systematic analysis of five generic dimensions across all CBM initiatives. This framework, called the CPI Framework², is designed for context analysis, processes evaluation and impact assessment of CBM initiatives. The CPI framework takes into account the multidimensional characteristics of CBMs and recognizes the difference between the aspects that are intrinsic to a CBM initiative and the contextual realities in which it is being or will be embedded, and with which it interacts. Adopting the CPI Framework enables a systematic evaluation of the contextual aspects related to participation, power dynamics, and technological settings. In this paper, we use this framework to showcase an in-depth qualitative empirical research into the contextual realities of two CBM initiatives in the Netherlands and Kenya. Both of these initiatives are part of Ground Truth 2.0, an EU-funded project that is setting up and validating six CBM initiatives in Europe and Africa³. Ground Truth 2.0 has a 'people-centered' approach towards establishing and scaling up CBM initiatives and aimed to set up sustainable CBM initiatives by involving three main categories of stakeholders; citizens, data aggregators/scientists and decision makers. The CBM case study in the Netherlands focuses on the issue of pluvial floods in 'Land van Heusden en Altena', while the Kenyan CBM aims to help balancing sustainable livelihoods and biodiversity management in the Maasai Mara ecosystem.

This paper is structured as follows. Section 2 presents the theoretical background of the research and introduces the selected dimensions and aspects of the CPI Framework for this research. The method used for

applying the framework for capturing the baseline situation in the two aforementioned CBM initiatives is then discussed in Section 3. The two case studies of this research are introduced in Section 4. Section 5 is dedicated to demonstrating the application of the framework by depicting the baseline situation for the two CBM initiatives. Finally, general as well as case-specific discussions, recommendations and conclusions are presented in Sections 6 and 7.

2. Theoretical background

The CPI Framework consists of five generic dimensions across all CBM initiatives, namely, 'goals & objectives', 'technology', 'participation', 'power dynamics' and 'results of the CBM initiative' (Gharesifard et al., 2019). The aspects discussed in the two dimensions of 'goals & objectives' and 'results of the CBM initiative' relate to the internal characteristics of the CBM initiative, while the 'technology', 'participation', and 'power dynamics' dimensions cover a wide range of contextual factors about the issue in focus of the CBM (e.g. the institutional and political context), the CBM initiative itself (e.g. its establishment mechanism), or both (e.g. communication paradigm about the issue, and within the CBM initiative). To be able to capture the baseline situation of the contextual settings in the cases in this research, we have used the elements of the CPI Framework that focus on the contextual aspects (see Fig. 1). Specifically, these elements include the context-related aspects of 'power dynamics', 'participation' and 'technology' dimensions. This section is dedicated to introducing the key context-related aspects of the CPI Framework.

2.1. Power dynamics dimension of CBM initiatives

The power dynamics dimension focuses on the question 'who controls and influences the CBM initiative and how?' Answering this question first and foremost requires an in-depth understanding of the current power dynamics among stakeholders and existing practices regarding the management of the issue in focus of the CBM initiative. Gharesifard et al. (2019) identified (1) the institutional and political context related to the issue in focus of a CBM initiative, (2) authority and power of different stakeholder, and (3) their access to, and control over, data as three context-related aspects that influence the power dynamics within a CBM initiative (see Fig. 1).

Institutional and political context includes the formal and informal rules and regulations and the extent of their implementation, as well as the roles and responsibilities of different stakeholders and the place of public participation in these processes. Authority and power of different stakeholder refers to their actual level of impact or influence on decision making processes. Finally, access to, and control over, data relates to existing regulations and procedures that determine the level of access to data for different stakeholders.

2.2. Participation dimension of CBM initiatives

This dimension of the framework focuses on the question 'who participates in the CBM initiative and how?' Gharesifard et al. (2019) introduces three context-related aspects that can help depict the participation dynamics before the establishment of a CBM initiative; (1) identifying the stakeholders who are included in, or excluded from, managing the water-related or environmental issue in focus of the CBM initiative, (2) studying the existing communication pattern among these stakeholders, and (3) understanding how these stakeholders interact with each other, while participating in the decision making processes (i.e. the communication and participation method).

Identifying the stakeholders who are currently involved in managing the water-related or environmental issues will enhance our understanding of included and excluded groups in decision making processes regarding these issues. This is particularly necessary for CBM initiatives that aim to move beyond data collection and to facilitate

¹ The two terms 'contextual factors' and 'circumstantial factors' have been used as synonyms in this paper and both refer to the circumstances that influence the establishment and functioning of a CBM initiative.

² Conceptual framework for examining the contextual setting, process evaluation and impact of a CBM initiative

³ <https://gt20.eu/>

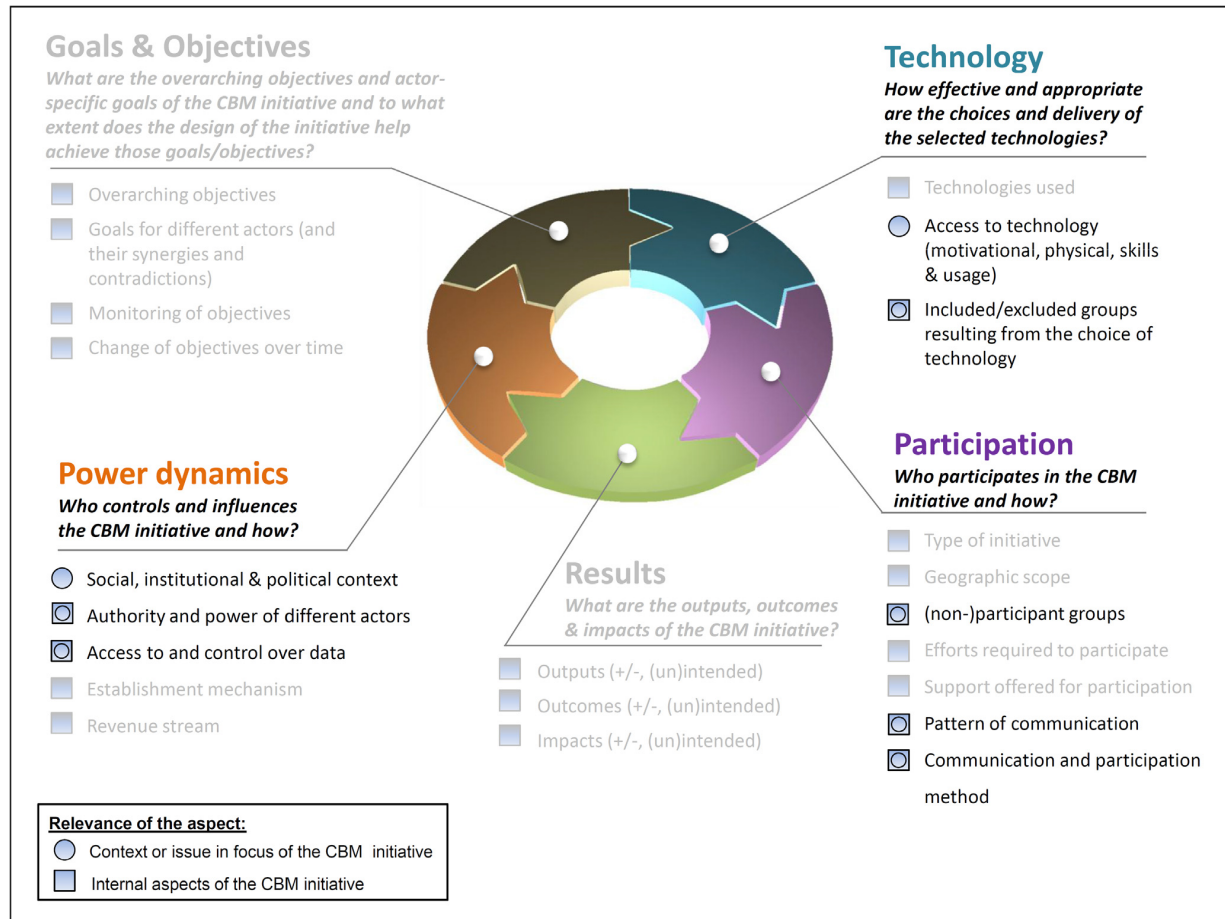


Fig. 1. Conceptual framework for examining the contextual settings of CBM initiatives (highlighted dimensions and aspects are context-related) Source: The CPI framework of Gharesifard et al. (2019), emphasis added.

public participation in decision making processes. Studying the existing pattern of communication and information flow among different stakeholders before the establishment of a CBM initiative is essential for explaining if and how a CBM initiative has mediated these interactions. Finally, studying existing methods of communication and participation in decision making processes is a prerequisite for understanding the extent to which an initiative may have affected or altered these interactions.

2.3. Technology dimension of CBM initiatives

Technology is an integral part of CBM initiatives and should enable participation of different stakeholders in a CBM initiative. Technology within a CBM initiative is either newly developed for the purpose of achieving the CBM goals (e.g. a newly developed App), or has previously existed and is adopted to provide desired functionalities within the CBM initiative (e.g. off the shelf air or water quality sensors). Technology can also be categorized based on its functionality within the CBM initiative (e.g. data collection, analysis or communication). The decision to select and adopt any (existing or newly introduced) technology comes with social, economic, political, and even cultural implications that need to be carefully considered. The key question for this dimension is therefore; 'how effective and appropriate are the chosen technologies?' Answering this question requires assessing the technological choices for a CBM initiative in relation to existing infrastructure as well as social and technological capabilities. This is in line with two

context-related aspects of the CPI framework, namely (1) the existing infrastructure and the different forms of access to technology (material, motivational, usage, and skills) among different stakeholder groups, and (2) included/excluded groups resulting from the existing technological settings (Gharesifard et al., 2019).

3. Methodology

The conceptual framework introduced in section 2 was developed within the scope of the Ground Truth 2.0 project. The results and discussions in this paper reflect the application of the CPI Framework, developed by Gharesifard et al. (2019), in depicting the baseline situation in two of the six Ground Truth 2.0 case studies, namely in Kenya and the Netherlands. The baseline situation in this paper refers to the contextual setting related to the issue in focus of each CBM initiative at their initiation.

For collecting the baseline data about the participation paradigms, power dynamics, and technological settings before establishment of the CBM initiatives, 60 in-depth semi-structured interviews were conducted in the Netherlands (26 interviews) and in Kenya (34 interviews) case. The complete lists of questions used for conducting the baseline interviews in the two cases were derived from the CPI framework and are provided as supplementary data in Appendix A. The average length of the interviews was 50 min and they were conducted either face to face or via phone/Skype. In general, four categories of stakeholders were approached for the interviews, namely participants in the co-design

meetings of the respective CBM initiatives, regulatory entities, members of the general public, and expert advisors (i.e. experts on the issue in focus of the CBM initiative). In many cases, the organizations and individuals that are categorized as regulatory entities or expert advisors were also present in the respective CBM co-design meetings and thus are also considered as members of the CBM co-design group. Because of the complexity of the topics and due to the fact that different stakeholders had different levels of knowledge or understanding about the issue in focus of the CBM and the processes that we were interested to study, we had to customize the interview questions for the four categories of interviewees (see [Appendix A](#) for further details). The sample from the general public was selected using snowball and stratified sampling (based on both gender and age), while interviewees from the other three categories were selected using cluster sampling. In case of the regulatory entities, we interviewed at least one respondent from the local or national authorities who has a formal or legal mandate for managing the issue in focus of the CBM initiative. Expert advisors were selected from the pool of individuals with knowledge or experience about the issue in focus of the CBM initiative who were either identified in the co-design meetings or were already known to the co-authors as a part of their professional network. As an extra sampling criterion, at least 60% of the participants in the co-design meetings were interviewed.

[Fig. 2](#) illustrates the detailed distribution of interviewees in the two cases based on their age, gender and stakeholder type. It is important to note that the age and gender distribution of the samples are influenced by factors such as the overall population composition of each case and the (age and gender) composition of the participants in the CBM co-design meetings.

The results and findings from the interviews were further complemented by a review of secondary sources as well as by the results of the discussions and observations made during several co-design meetings in the cases. The secondary sources include scientific publications, major laws, regulations, acts and statistics related to the issues in focus of the two CBM initiatives.

4. Case studies

The two CBM initiatives that were selected as the case studies for this paper are part of the Ground Truth 2.0 project. The CBM initiative in the Netherlands focuses on the issue of pluvial flooding in 'Land van Heusden en Altena', while the CBM initiative in Kenya focuses on a much wider set of environmental issues that affect biodiversity conservation and livelihood management in the Maasai Mara ecosystem. The selection of these two cases enabled conducting a rare comparative study of the factors that influences the establishment and functioning of CBM initiatives in the context of a developed country versus a developing country. Although the difference in the thematic foci of the cases limited the possibility of a direct comparison of some of the aspects, it allowed for demonstrating the applicability of the CPI framework for conducting a baseline analysis of CBM initiatives regardless of their aims and objectives. In addition, it helped strengthen the overarching message of this research (i.e. the need for careful consideration of contextual realities while establishing CBM initiatives), which cuts across initiatives with different thematic foci.

4.1. The Dutch case study: Grip op Water Altena

The Dutch case study is located in the 'Land van Heusden en Altena' which is a part of the Dutch province of North-Brabant. In terms of water management, this area falls under the jurisdiction of the Rivierenland Regional Water Authority and consists of the three municipalities of Werkendam, Woudrichem and Aalburg. With an increase in the number of intense rainfall incidents in recent years (e.g. two extreme rainfall events in the summers of 2014 and 2015), pluvial flooding has become a major concern and has negatively affected a

number of residents in this area.

The agreed goal of the stakeholders in this CBM initiative is therefore to prevent damage from extreme precipitation and this is reflected in the name of the initiative, 'Grip op water Altena' (in Dutch), which means grip on water. This will be achieved by improving the status quo of communication and data, information and knowledge sharing between local citizens, researchers, three aforementioned municipalities and the Rivierenland Regional Water Authority. The CBM initiative in this case study is facilitated by a web-platform⁴. This platform is used by local stakeholders to raise awareness and communicate about the measures that they have taken to improve water retention capacity in the area of Land van Heusden en Altena. For example, the local authorities utilize this platform to communicate about newly constructed or improved water infrastructure or water retention areas and citizens can share information and tips about their local measures, such as removing extra tiles from their garden or attaching their roof downspout to a rain barrel.

4.2. The Kenyan case study: Maasai Mara Citizen Observatory

The Kenya case is located in Narok County in southwestern Kenya, close to the Tanzanian border. This area includes the Maasai Mara National Reserve, the Mara Triangle and conservancies around the national reserve. This is a part of the wider Mara ecosystem and is being managed by the Narok County government. The majority of the inhabitants of this area are Maasai pastoralists and it is one of the most famous touristic destinations in Kenya. The human-wildlife conflict is a prominent issue in this area and practices such as overstocking, overgrazing and fencing alongside droughts have put a lot of pressure on biodiversity and people's livelihoods.

The Ground Truth 2.0 CBM initiative in Kenya is called the 'Maasai Mara Citizen Observatory' (MMCO) and aims to support balancing sustainable livelihoods and biodiversity management in the wider Mara region. This will be achieved with the help of citizens and key stakeholders involved in managing biodiversity and livelihoods in the Mara region. MMCO is supported by a web-platform⁵ and an App that enable the users (including local community members, tourists, tour guides, researchers, rangers and local authorities) to collect, share and receive data about a wide range of topics including biodiversity sightings, natural hazards (e.g. floods and droughts), pollution and emergency situations. In addition, the Ground Truth 2.0 will install a number of inexpensive (TAHMO⁶) weather stations in the project area, which will collect meteorological data and share this via the MMCO web-platform and App. This data will help the local communities to have a better overview of available water resources and grazing lands and the authorities to predict species' distributions. The key stakeholders who were involved in the co-design process of the Kenyan CBM case included local community member, scientists and researchers from local and international research institutes, NGOs, conservancies, and representatives of the Narok County and national government.

5. Baseline situation in the two case studies

This section depicts the baseline situation in which Grip op Water and the Maasai Mara Citizen Observatory were being established. The findings are organized based on the eight key context-related aspects across the 'power dynamics', 'participation' and 'technology' dimensions of the CPI framework (see [Fig. 1](#)). Each sub-section (i.e. 5.1, 5.2, etc.) focuses on one of these aspects and starts with a cross-case comparison of the findings. The case-specific findings of the two case studies are then presented under each sub-section.

⁴ <http://altena.gripopwater.nl/>

⁵ <https://mara.info.ke/>

⁶ <https://tahmo.org/>

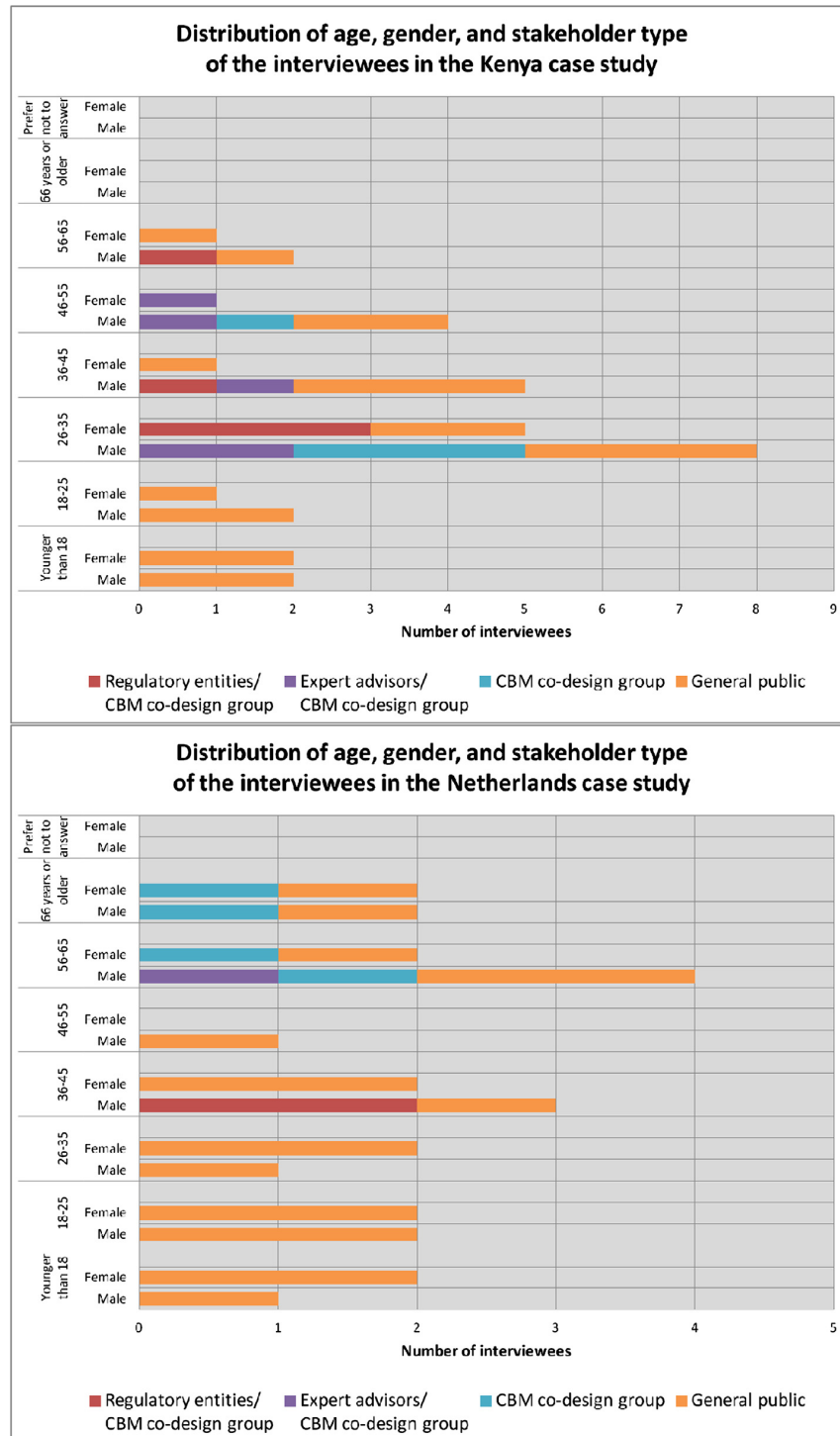


Fig. 2. Age, gender and stakeholder type of the interviewees in the two cases.

5.1. Power dynamics dimension: Institutional & political context

The institutional and political context in the Dutch case study can be characterized as a well-established decentralized water governance system that has increasing attention to the issue of pluvial flooding in urban areas. The stakeholders describe this as a well-functioning

system, in which citizens have a high level of trust in the authorities responsible for water management; however, an awareness gap exists among the citizens about the topic of water management. In contrast, the Kenyan case study has a recently devolved system of water and environmental governance; a young system which still heavily relies on the central government in terms of policies, procedures and resources.

The level of trust in the authorities responsible for the management of natural resources is much lower in the Kenyan than in the Dutch case study and the local communities are more concerned and conscious about the affect of environmental management and conservation practices on their daily lives.

5.1.1. Institutional & political context in the Dutch case study

There are three major institutional layers in the Dutch water management system; (1) European level, including the EU and International River Basin Commissions; (2) national level, which includes the central government (Ministry of Infrastructure and the Environment) and the national Water Authority (Rijkswaterstaat); and (3) sub-national level, which consists of the 'triangle' of provinces, Regional Water Authorities and municipalities (OECD, 2014). Dutch water management has been traditionally decentralized from both, functional and territorial perspectives (OECD, 2014). Functional decentralization refers to the tasks that each decentralized unit carries, while territorial decentralization is linked to different provinces and municipalities within the country. In terms of hierarchy, the Regional Water Authorities and municipalities are both at the same level, while the provinces have a supervisory function over both. The provinces are also responsible for laying down rules about water management and setting up or dissolving Regional Water Authorities (Havekes et al., 2017).

During the past eight years, the issue of pluvial flooding has moved from being absent in the Dutch water policy for flood protection, climate adaptation and water resilience to being high on the political agenda in Delta Programmes 2016, 2017 and 2018. In practice, the main actors involved in managing pluvial flooding are the municipalities and Regional Water Authorities. This is due to the fact that the *Water Act (2010)* places the municipalities in charge of processing 'urban waste water', which includes rainwater run-off, in the municipal area and obliges the municipalities and Regional Water Authorities to coordinate their activities. The issue of pluvial flooding in urban areas is closely linked to spatial planning. Municipalities and property developers are the main stakeholders who benefit from spatial development in the Netherlands; nevertheless, unlike the Regional Water Authorities, they do not bear much of the cost when it comes to water management (OECD, 2014), nor is water management among their primary tasks. Moreover, the Regional Water Authorities are independent from the central government in terms of generating their own revenue, while municipalities are largely dependent on the national government for their income via municipal funds.

In addition to the *Water Act (2010)* and the Delta Programmes, there are several other laws, agreements, plans and policy guidelines related to the topic of this CBM initiative, among which are the National Administrative Agreement on Water, the Spatial Planning Act, the National Water Plan 2016–2021, the Noord-Brabant Provincial Environment and Water Plan 2016–2021, the water management plan of the Rivierenland Regional Water Authority, the sewerage policy vision of the Werkendam, Woudrichem and Aalburg municipalities and the common water plan of the municipalities and the Rivierenland Regional Water Authority. Moreover, the Dutch government is currently working on developing the 'Environmental Planning Act' (expected to take effect in 2021); an Act that will combine several existing laws, including the Water Act, the Crisis and Recovery Act and the Spatial Planning Act into one simplified and coherent piece of legislation.

Almost all interviewees believed that the implementation of the rules, roles and responsibilities related to the management of pluvial floods is very strict and in some cases even stricter than the available protocols. Several interviewees considered the fact that they do not see much of a problem with pluvial floods in the Netherlands as a sign of good implementation. Moreover, a number of interviewees mentioned that they are satisfied with the fact that there are numerous agreements, protocols and procedures in place and being implemented, especially in emergency situations.

The majority of the interviewees argued that the authorities are responsible for managing issues such as pluvial flooding and that civilians pay for this service via taxes. The interviewees also believed that trust in the government is deeply rooted in Dutch culture and that Dutch citizens do not want to think about such problems too much. Several interviewees from all stakeholder groups mentioned the Dutch quickly shift attention to matters visible for them. For example, in case of an incident (e.g. a major flood), the Dutch community tends to come together and help voluntarily to prevent loss of lives or damage to properties. The result of the interviews also revealed an 'awareness gap' among the Dutch citizens regarding water management, an issue that was also highlighted in OECD (2014). This awareness gap, combined with high level of trust in the government, can hinder the implementation of necessary actions in the prevention phase of an infrequently experienced problem such as pluvial flooding.

5.1.2. Institutional & political context in the Kenyan case study

Given the fundamental changes in the Constitution of Kenya (CoK) in August 2010, the country's institutional framework took a major turn from a centralized system of administration and governance to a devolved structure of government which cedes power of legislation, execution and also revenue collection and expenditure to 47 counties. The judiciary power, however, remained central at the national level. With these changes, the new constitution introduced two main levels of government: national and county. In terms of hierarchy, the constitution defines these two levels as parallel and does not indicate any superiority for one level over the other (Government of Kenya, 2010-Article 1), but several Articles emphasize the need for consultation collaboration and coordination between the two levels. For example, Article 6 (2) defines these levels as 'distinct and inter-dependent', but highlights the need for their 'mutual relations on the basis of consultation and cooperation'. Two main points were elicited from the interviews. Firstly, the county is autonomous, but its decisions should not conflict with the national (parliament) law and in any conflicting case, the national law prevails. Secondly, a number of interviewees believed that according to the devolution of power, the county government has the final decision making power; yet in many cases it is still the national government that takes some of the measures on the ground.

Apart from the CoK, the interviewees identified (inter)national-level legislation and policy guidelines related to the topic of the CBM, including the agreement for the joint trans-boundary management of the Mara River Basin, the Kenyan law on the management of protected areas, the Wildlife Act, the Water Act 2005, and the Forest Act. At the county level, the Narok County Integrated Development Plan and the Environmental Management Bill are the main pieces of legislation and, for many other aspects, the county uses bylaws driven by the national policies and customized to the local issue. Nevertheless, at the time of conducting this research, the Narok County Government was in the process of approving a number of bills that will substitute these bylaws, namely the Maasai Mara Management Plan, the Narok County Tourism Act, the Environmental Management Act and the Livestock Act.

The vast majority of the interviewees believed that the extent of implementation of rules, roles and responsibilities regarding biodiversity conservation and livelihood management in Kenya highly depends on the specific issue, location and actors involved, but, overall, implementation is not very strict at both, local and national levels. Interviewees often referred to increasing wildlife deterioration and the existence of multiple barriers for sustaining livelihoods to back up their claim. They identified several institutional, social and political barriers for improved implementation, including lack of resources (money, knowledge and staff); especially at the county level, conflicting personal and political interests and corruption. Moreover, it was mentioned that strengthening implementation by the county is difficult, because the county does not have the judiciary power to enforce the law; for example, in a poaching incident, only the Kenya Wildlife

Service (KWS) can arrest and prosecute people; if the county rangers want to arrest someone, they have to report it to KWS. The same applies to the issue of illegal logging and the Kenya Forest Service (KFS).

Interviewees also highlighted several values, norms and traditions of the Maasai culture that can positively or negatively influence biodiversity conservation and livelihood management in the Mara. For example, the traditional land ownership model of the Maasai was community land ownership instead of individual land ownership. Preserving this tradition can help stop controversial practices such as fencing, which increasingly cause wildlife deaths. On the other hand, Maasai are traditionally pastoralists, and they consider the number of livestock they have as an indicator of their wealth; therefore, overstocking and overgrazing have become prominent issues in the Mara region.

5.2. Power dynamics dimension: Authority and power

There was not a big difference in the perceived level of authority and power among the stakeholders in the two cases. The regulatory entities in both cases are the most influential stakeholders in decision making processes. They either have direct authority or a mandate for management of water or environment-related issues or they participate in co-governing the resources with other stakeholders, or, alternatively, provide them with advice or consultation. In contrast, the majority of the general public and expert advisors in both cases perceived no direct influence on decision making processes or that they only have an indirect influence through participation in elections, awareness raising or via communicating with authorities to express their concerns. There is, however, one clear difference between the two cases: whereas in the Kenya case, the roles of stakeholders are perceived to be strongly gendered, this was not the case in the Dutch case.

5.2.1. Authority and power in the Dutch case study

All interviewees from regulatory entities in this case mentioned that they exert influence on decisions related to pluvial flood management by providing advice/consultation, joining other colleagues or other organizations for decision making, or by having direct authority on the decisions because of their job. Expert advisors who are not part of regulatory entities mainly have an influence by providing advice to the authorities, or communicative influence, for example by publishing their research findings.

The majority of the interviewees from the CBM co-design group and the general public perceived to have little or no influence on decisions related to managing pluvial floods, or mentioned that they have an indirect influence via participation in the elections of Regional Water Authorities. Several participants also mentioned they do not (automatically) receive information about such decisions; however, if they are interested, they can search for and find this information. On the other hand, a few interviewees from the CBM co-design group and the general public gave examples of communicative influence on decisions, such as signing a petition, writing a letter to the authorities and expressing their opinions on how to manage pluvial floods, or campaigning via social and mass media. Regarding the latter, one of the interviewees argued that “you can always complain by publishing an article in a newspaper or writing a note on their Facebook page; ‘public shaming’ works”.

5.2.2. Authority and power in the Kenyan case study

All interviewees from the regulatory entities at the county and national levels perceived direct influence on decisions related to biodiversity conservation and sustainable livelihood management because of their official mandate. According to the CoK and a number of interviewees from the local regulatory entities, the county government has the strongest influence among all actors. Nevertheless, local chiefs play a vital role in the balance of power between the national and local government. They have an explicit influence on daily local decisions

and act as a direct link between the community and national authorities. They have an advising/consulting role towards the national government and relay the decisions and agreements that are made at the national level to the community. Moreover, the church was also identified as one of the most powerful stakeholders; a platform which can communicate directly with the authorities and influence public opinion. In this regard, a spiritual leader from a church said: “we provide advice and consultation to the authorities; even the president sometimes consults the church on different decisions”.

The respondents from the CBM co-design group, the expert advisors and the general public expressed two ways in which they can have a (minimal) influence on decisions: (1) via altering public opinion (rather than policies and decisions) and educating community members (e.g. via awareness raising programs); or (2) by providing advice to the authorities inside or outside of participatory meetings. The rest of the interviewees from these groups stated that they have little or no influence on decisions and they are informed about decisions when these are gazetted or advertised via other channels. In this regard, one interviewee mentioned: “our chance to influence the decisions is very limited; they [the authorities] are not ready to listen to just a simple person”.

Some of the respondents from the general public also emphasized that at household level, the decision making power normally rests with men; female members of the communities are less involved in making decisions.

5.3. Power dynamics dimension: Access to and control over data

Both the Dutch and the Kenyan constitutions explicitly express the right of access to information for the general public; nevertheless, the interpretation and implementation of this issue differs between the two cases. In the Dutch case, public authorities have explicit deadlines for responding to requests for data and in case of denying access, they need to clearly justify their choice. In the Kenyan case, however, a rather open interpretation of ‘sensitive’ or ‘exempt information’ seems to be used sometimes to deny access to data sharing requests. Moreover, in practice, in both cases, access to data is often controlled by the organizations or individuals who collect or hold the data.

5.3.1. Access to and control over data in the Dutch case study

In line with Article 110 of the Constitution of the Kingdom of the Netherlands, all Dutch government information has to be shared with the public and any information that is not already publicly shared (e.g. via websites) can be requested in accordance with the Dutch Public Access to Government Information Act ([Law of 31 October 1991](#)). This law, however, also allows government organizations to withhold certain information if considered confidential for privacy or security reasons. Public authorities have explicit deadlines for responding to requests, and in case of refusal should provide the applicants with the reasons for their decision.

The interviewees from the regulatory entities and the expert advisors believed that a lot of data is available about pluvial floods, but the general public is not aware of this because the information is mostly stored in internal databases within organizations and thus not immediately available for the public. The majority of the remaining interviewees, especially those from the general public, did not have any experience with accessing data about pluvial floods; however, the few who did, mentioned that some organizations, e.g. the Dutch Meteorological Institute (KNMI), make more data publicly available than others.

The interviewees identified a wide range of organizations and individuals as qualified to analyze data about pluvial floods, including the Ministry of Infrastructure and the Environment, Rijkswaterstaat, the Regional Water Authorities, the Union of Regional Water Authorities, provinces, municipalities, KNMI, NGOs, consultancies, scientists and scientific institutes. Only a few interviewees considered citizens among

this group.

The Regional Water Authorities were identified as the main organization that collects data about pluvial floods at the local level and Rijkswaterstaat was identified as the main producer and owner of data at the national level. In addition, some interviewees believed that municipalities and provinces hardly take measurements themselves, and even if they do, the measurements are mostly qualitative (e.g. whether or not a sewage overflow occurred), not quantitative (e.g. how much sewage water has been released to the surface waters). Regarding control over data, it was mentioned that it is mainly the owner of the data who defines the level of access for others.

5.3.2. Access to and control over data in the Kenyan case study

Article 35 of the CoK (2010) explicitly indicates that every Kenyan citizen has the right of access to the information held by the State, or by another person, if this is required for exercising or protecting any right or fundamental freedom. Furthermore, the state shall publish and publicize any important information affecting the nation. With reference to this Article, the [Access to Information Act \(2016\)](#) and Part IX of the [County Government Act \(2012\)](#) preserve the right of Kenyan citizens for requesting access to information held by any government organization, including the county government or any unit or department thereof. Despite these written rules, a great majority of the interviewees believed that the level of access to data actually depends on the organization, department or individuals who hold the data. Several interviewees criticized the current (limited) data sharing practices of the authorities, based on their previous experiences and encounters. For example, a respondent mentioned “sometimes they give you the data, and sometimes they simply decide not to”; another one argued that “sometimes you will be asked why you need the data in a harsh way; they have the fear of use of this information by the community”. This is due to the fact that certain data and information are deemed ‘sensitive’ or ‘exempt information’ and although defined in the legislation, the definition is open to interpretation. This has resulted in keeping essential information from the public. Moreover, some expert advisors stated that, in any case, one should not publish data that is not accepted or endorsed by the government; otherwise they will advertise against using the data and claim that the data is not good at all.

All interviewees unanimously believed that the availability of data about biodiversity and livelihoods is not very good. The available data was often described as inaccurate, unreliable, not up-to-date, not well distributed (spatially and temporally), and inaccessible; especially publicly or in a central way. Furthermore, lack of coordination between organizations in terms of collecting and sharing data was identified as another issue. It was mentioned that if data exists, it is difficult to know who has the data and the organizations who hold the data do not share it with the public and sometimes not even with each other.

Research organizations, individual experts/scientists, NGOs and government organizations were identified by the interviewees as the best-placed stakeholders for analyzing data on biodiversity and livelihoods; however, it was mentioned that for some data, the analysis requires highly specific expertise that may not even exist among the county rangers. On the other hand, many interviewees believed that the local people have a very good understanding of biodiversity and livelihoods, but literacy and language are barriers that affect their ability to share their knowledge.

5.4. Participation dimension: (Non) participant groups

Although public participation has been gaining more attention in the Dutch case study, it is not a fundamental pillar of Dutch democracy and not an integral part of current water management practices. In contrast, public participation is considered a governance principle in the Kenyan case study and has a vivid footprint in several pieces of legislation. Nevertheless, examining the main stakeholders who are currently involved in the management of the respective issue in focus of

these CBM initiatives revealed that despite the legislative differences, the role of the general public in both cases is minimal, often limited to electing representatives.

5.4.1. (Non) participant groups in the Dutch case study

The interviewees were asked to identify those stakeholders who they think are currently involved in decision making and policy making processes regarding the management of pluvial floods in the Netherlands. The identified stakeholders included the major actors involved in water management in the Netherlands, namely the Regional Water Authorities, municipalities, Provinces, Rijkswaterstaat, Ministry of Infrastructure and the Environment, Union of Regional Water Authorities (Unie van Waterschappen), the Association of Dutch municipalities (Vereniging van Nederlandse Gemeenten). Moreover, the interviewees also identified the Rioned Foundation, an umbrella organization for urban water management and sewerage in the Netherlands, and the Southern Agriculture and Horticulture Organization (ZLTO), which is a farmers association with 15,000 members in Zeeland, Noord-Brabant and Zuid-Gelderland. In addition, the interviewees also perceived an advising role for research and consulting companies from the private sector in managing pluvial floods.

The role of the citizens, however, was mentioned to be limited to electing representatives in the parliamentary, provincial, municipal, or Regional Water Authority elections. This is due to the fact that public participation is not a fundamental pillar of Dutch democracy. There is no mention whatsoever of public participation in the Constitution of the Kingdom of the Netherlands, nor in the [Water Act \(2010\)](#). Although citizen participation is gaining more attention in more recent policy guidelines such as the National Water Plan 2016–2021, fundamentally, it is “at best, as an instrument to improve the current working of representative democracy” (Michels, 2006, p. 336).

5.4.2. (Non) participant groups in the Kenyan case study

The Parliament of Kenya was identified as the main body of policy making regarding biodiversity conservation and livelihood management at the national level. Cabinet secretaries, especially those who head the ministries, involved in managing biodiversity and livelihoods, relevant wings of ministries such as KWS, KFS, Kenya Water Resources Management Authority (WRMA), the National Museums of Kenya and the Kenya Water Tower Agency were identified as the most relevant actors for the issue in focus of the CBM initiative at the national level. Furthermore, the Narok County Government was elicited as the most influential stakeholder at the local level. Local chiefs, conservancies and NGOs were also deemed relevant stakeholders at the local level. Despite the devolution of power from the national level to the county government in Kenya in 2010, the national level organizations such as KWS, are still very much involved in conservation activities at the county level.

Public participation has a vivid footprint throughout the CoK (2010) and in several other pieces of legislation including the [County Government Act \(2012\)](#), the [Public Service \(Values and Principles\) Act \(2015\)](#), and more specifically related to the topic of this CBM initiative, the [Environmental management and Coordination Act \(1999\)](#), the [Wildlife Conservation and Management Act \(2013\)](#) and the [Agriculture and Food Authority Act \(2013\)](#). The first Article of the constitution portrays public participation as a right for every Kenyan and states that all sovereign power belongs to the people of Kenya and that they can exercise their power directly or through their elected representatives. The Constitution also recognizes participation of the Kenyan people as a national value and a governance principle and obliges the state to facilitate participation of everyone in governance processes. Furthermore, public participation is also mentioned as a purpose of devolution of government for providing self-governance power to the people and to enhance their participation in decision making processes. The national and county governments are obliged to facilitate and encourage public participation in the legislative process, public finance and also

conservation of environmental and natural resources management.

Despite the formal anchoring of participation in these legal frameworks, none of the interviewees perceived any involvement of the general public in policy and decision making processes regarding biodiversity conservation and livelihood management. This, among other things, presents a picture of the limited extent of implementation of these legislations. Despite much emphasis on public participation in the CoK and the aforementioned acts, Kenya still lacks a national act that provides a general framework for how to give effect to the constitutional and legislative requirements in this regard. It has been argued that such a framework at the national level is needed to guide the public participation processes at the local level (Marine, 2015). The [Public Participation Bill \(2016\)](#) was prepared to cover this gap; however, this bill still has not been accepted due to controversies and opposing ideas. At the time of conducting this research, the [County Public Participation Guidelines \(2016\)](#) was the only available (but not legally binding) tool to guide this process.

5.5. Participation dimension: Communication paradigm

Although pluvial flooding is a popular conversation topic among the Dutch, the predominant pattern of communication about this topic is unidirectional, from authorities to the citizens. Citizens only communicate about pluvial flooding with authorities, if there is an acute problem or an emergency situation. In such situations, people would like to receive a quick response and therefore mostly communicate using phone calls. Similarly, in the Kenyan case study, the authorities mostly communicated about biodiversity conservation and livelihood management unidirectionally and for the purpose of informing citizens.

5.5.1. Communication paradigm in the Dutch case study

The [Safety Regions Act \(2010\)](#) sets the legal framework for risk and crisis communication in the Netherlands. This Act divides the country into 25 security regions which are extended local government units with joint safety regulations. The management board of a security region is responsible for informing a number of government authorities and personnel as well as the citizens about risks and crises that may affect a specific region. Although occasional bi-directional and interactive patterns of information flow for risk and disaster communication can be found within and between (central) government authorities, the predominant pattern of information flow is unidirectional, from authorities to the citizens. In addition, this Act only focuses on such communication at regional level; however, as [Kaufmann et al. \(2016\)](#) clarified, communication of preparatory and responsive measures for events such as pluvial floods at a smaller scale can highly differ from one location to another.

The interviewees were asked whether or not they communicate about pluvial flooding with others and to indicate their preferred channels for such communication. Using Apps on smart phones (e.g. WhatsApp, Viber, Line) and communicating via email, telephone (call or SMS), or face to face (e.g. daily bilateral discussions with others or participation in meetings) were among the most frequent means of communication in this case study. Interviewees from the regulatory entities and expert advisors indicated that they communicate about this issue mostly for work purposes, both internally (e.g. with colleagues) and externally (with other individuals and organizations e.g. with residents or Regional Water Authorities). Interviewees in this case often linked the issue of pluvial flooding with bad weather and intense rainfall incidents. The interviewees from the CBM co-design group and the general public mentioned they often communicate about the issue with other residents in informal settings or after heavy rainfall. At the same time, a number of interviewees from these groups emphasized that unless there is an actual problem or an emergency situation, there is no or too little communication with municipalities and other authorities. [Fig. 3](#) summarizes the preferred channels for communicating. Based on this figure, the most preferred channel for communication was

phone call or SMS, as people prefer to receive a quick response or like someone listen to them.

5.5.2. Communication paradigm in the Kenyan case study

There is no article in the CoK or other legislation that directly mentions the desired pattern of information flow between different stakeholders for policy and decision making processes; however, several articles portray a 'unidirectional' communication paradigm as the prescribed pattern of communication from state and county government to the general public. The main purpose seems to be informing the public about decisions. As an example, the [County Government Act \(2012\)](#) requests county governments to integrate communication in all development activities.

Using telephone and smart phone App calls (predominantly WhatsApp) are by far the two most frequently used means of communication in this case study. Due to the low average level of literacy among local citizens, phone calls are much more common than sending SMS. Different communication channels are used for reaching different target groups; for example, social media (e.g. Facebook) is one of the best channels for reaching out to the youth, while for elders face to face communication works better. Moreover, radio was indicated as an efficient channel for reaching a large number of people, even in remote areas. The majority of the interviewees mentioned that they communicate about biodiversity conservation and livelihood management both for work and personal purposes. The authorities mainly prefer to communicate with community member face to face, while the community members mostly use face to face meetings, phone calls and SMS, WhatsApp and email for reaching out to the authorities. [Fig. 4](#) presents the preferred channels for communication. As the Figure shows, using an App on a smartphone (predominantly WhatsApp) and using websites or blogs are, respectively, the most and least preferred channels for communicating about biodiversity conservation and livelihood management in this case study.

5.6. Participation dimension: Methods of communication and participation

In both cases, decision making about the issue in focus of the CBM initiatives has a top-down form and relies heavily on the technical expertise of the regulatory entities and sometimes involves deliberation and negotiation between the stakeholders in charge of managing the issue. Although several respondents from the expert advisors, members of the co-design group and the general public in the two case studies perceived to have little or no influence on decision making processes, respondents from the Kenya case study indicated a much wider range of methods by which they could lobby or communicate their ideas and interests to the authorities and possibly intervene in decision making processes.

5.6.1. Methods of communication and participation in the Dutch case study

The interviewees were asked to explain in what ways (if any) they are involved in managing pluvial floods and how they take part in decision making processes on this issue. All interviewees from the regulatory entities mentioned that they are involved in managing pluvial floods via their jobs. Their involvement spanned from setting policies at the Regional Water Authority or municipal level to decision making at the operational level (e.g. managing the pumps and weirs for preventing floods). The examples provided by the interviewees from this group depicted the 'deliberate and negotiate' or 'technical expertise' roles for them in the decision making processes ([Fung, 2006](#)).

The expert advisors are mostly involved indirectly via providing data and information, as well as expressing their expert opinion in meetings with the authorities. These data, information and expert views may or may not be used by the authorities. Thus they are mostly involved in decision making processes via explicit data provision and expressing preferences.

The majority of the interviewees from the CBM co-design group and

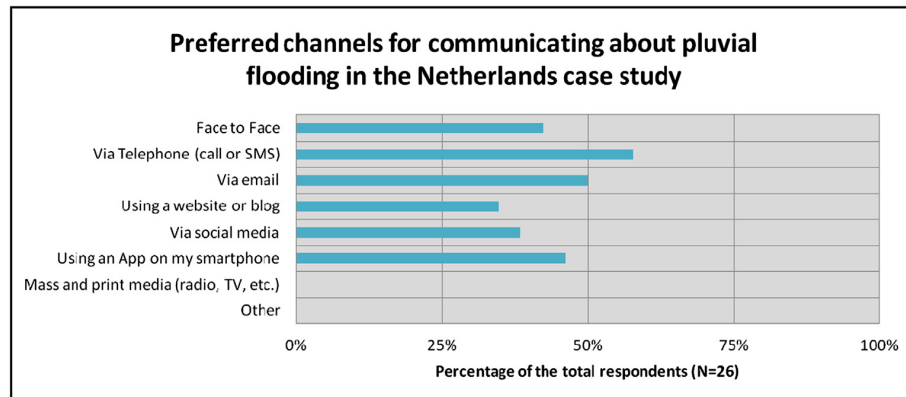


Fig. 3. Preferred channels for communicating about pluvial flooding in the Netherlands case study.

the general public had little or no expectation to influence decisions at any level. In this regard, one of the interviewees mentioned that “I don’t think I have any influence, it is the municipality who makes the decisions. If I have a problem, I contact the municipality and they decide how to take care of the issue”. A few interviewees, however, believed that they have an indirect say in decisions by participating in the elections of the Regional Water Authority elections, but they did not consider this as ‘real’ involvement in making decisions or creating policies.

5.6.2. Methods of communication and participation in the Kenyan case study

In the Kenyan case study, all participants from the regulatory entities perceived they have a direct or indirect say in the decisions because of their technical expertise. The expert advisors indicated that they are mostly involved via their jobs and the nature of their involvement was highly dependent on their work. This level of involvement ranged from expressing opinions by communicating about their research, to deliberation and negotiation with the authorities (see Fung, 2006).

Many interviewees from the CBM co-design group and the general public perceived that they are barely or not at all part of decision making processes related to biodiversity conservation and livelihood management. Some of these interviewees indicated they only get to know about decisions when these are gazetted. The other interviewees from the CBM co-design group and the general public identified two main methods by which they are involved in decision making processes. The first method was expressing preferences in participatory meetings or via membership in community action groups. Although it is not

always possible for everyone or even the community representatives to participate in such meetings (because of the location or timing of the meetings), those who do participate have the opportunity to express their preferences and even bargain with the authorities. For example, a local Maasai gave a clear example of bargaining power that he had because of his membership in a local water users association: “When we participate in these meetings, we express our opinion and contribute to the discussions; for example, through appealing to the 2010 constitution and the recent Wildlife Act we could get them [the Narok County Government] to assign more compensation for livestock deaths”. The second identified method was by communicating with different (influential) stakeholders bilaterally outside of participatory meetings. Interviewees highlighted that using this method depends heavily on social or professional links with these stakeholders. In this regard, one interviewee mentioned “if you want to have a good influence, you should be present in the political forums, or know people who are there”.

5.7. Technology dimension: Access to technology

The issue of access to technology in the two case studies of this research presents a classic North-South or ‘developed versus developing’ country situation. There is a drastic difference between the Netherlands and Kenya in terms of both, diffusion and actual use of ICTs; a difference that is amplified by the low level of literacy and required skills for effective use of ICTs in the Kenyan context. Moreover, in terms of ICT infrastructure, Narok County is largely under-developed as compared to the national average level.

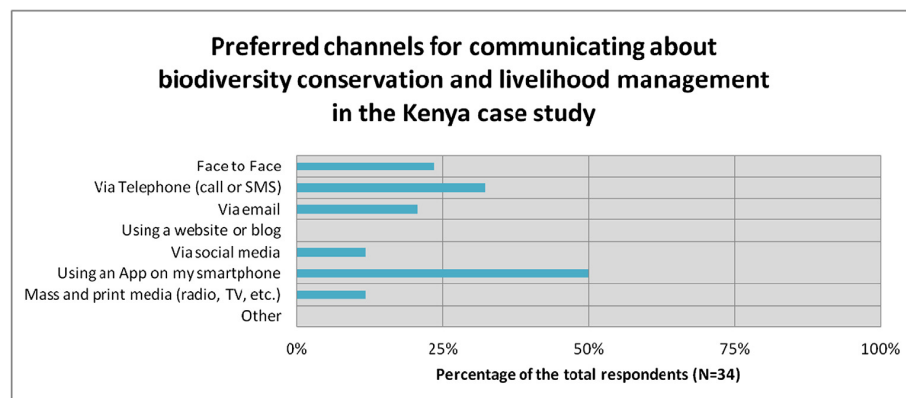


Fig. 4. Preferred channels for communicating about biodiversity conservation and livelihood management in the Kenya case study.

5.7.1. Access to technology in the Dutch case study

At the time of conducting this study, with an IDI rank 7 (ITU, 2017a) and a Digital Economy and Society Index (DESI) 4 (European Commission, 2017), the Netherlands ranked high in terms of ICT development in Europe as well as worldwide. Fulfilling the growing demand for high-speed Internet access is at the forefront of Dutch government policy (ITU, 2017b). In 2017, there were 129.9 mobile-cellular telephone subscriptions and 40.3 fixed-telephone subscriptions per 100 inhabitants in the Netherlands. Moreover, 92.2% of households had internet access and almost 90% of households had computers.

The country also ranked high in terms of actual use of information and communication technologies, 90.4% of citizens used the internet. Dutch citizens also have the required skills to use ICTs. According to the ITU (2017a), the 'mean years of schooling' in the Netherlands is 11.9 years. Moreover, the Dutch government aims to make digital literacy a core part of the curriculum of primary and secondary education (Ministry of Economic Affairs, 2016).

5.7.2. Access to technology in the Kenyan case study

Mobile phones are the main medium for digital communication in Kenya and at the time of conducting this research there were 81.3 postpaid and 'active' prepaid mobile-cellular telephone subscriptions per 100 inhabitants in Kenya (ITU, 2017b). In contrast, only 0.15% of residents had fixed-telephone subscriptions, less than 15% of households had computers and 22.3% of households had internet access.

Although a large proportion of the population used mobile phones, only some of these individuals were (actively) using the internet (World Economic Forum, 2016). The average low level of literacy in Kenya is a challenge for developing the required skills for effective use of ICTs. In 2017, the 'mean years of schooling' in this country was only 6.3 years.

In terms of both, access to and the skills to use ICTs, Narok County was performing below the national level. The strategy for spatial development of ICT infrastructure in Kenya is very much focused on service provision for population centers. For example, in 2016, with a geographical coverage of only 17%, up to 78% of the Kenyans had access to 3G broadband service (Intelecon, 2016). The downside of this strategy is that a large part of the counties, including a large proportion of the Narok County that has a dispersed population, are either underserved (e.g. do not have access to 3G broadband) or not served at all. Moreover, a recent report by the Kenya National Bureau of Statistics indicates that only 11% of Narok County inhabitants have a secondary level of education and 38% of the residents in this county have no formal education (Ngugi et al., 2013).

5.8. Technology dimension: Included/excluded groups resulting from technological choices

In addition to differences in access to technology for different groups within society, lack of basic digital skills by a considerable proportion of the population in both cases and a low average level of literacy among Kenyans are among the factors that may contribute to exclusion of a part of the population in processes that require using ICTs.

5.8.1. Included/excluded groups resulting from technological choices in the Dutch case study

Despite a well-developed and growing ICT infrastructure and availability of affordable mobile-cellular and mobile-broadband offers (ITU, 2017b), at the time of conducting this research, the access gap was not yet closed in the Netherlands. The results of a recent study of 108,000 Dutch citizens reveal that the younger population, with higher levels of education and income and (in some areas) males have better access to internet (van Deursen & van Dijk, 2015). In terms of actual usage, according to available statistics, a very large proportion of the Dutch population is active technology users. Nevertheless, the main gap resulting from the growing digitization of the Dutch society seems to

exist for those who lack basic digital skills. According to the European Commission (2017), this is no less than 23% of the Dutch population aged 16–74; a thought provoking figure but one that is still much lower than the 44% average for the whole Europe.

5.8.2. Included/excluded groups resulting from technological choices in the Kenyan case study

Despite advancements in the ICT sector in Kenya, there is still a big gap in both, access and skills to effectively use technology. In addition to differences in accessing technology, the low level of literacy (especially among the older generation) and gender discrepancies in access to and use of technology are among contributing factors in creating new forms of inequalities (Brännström, 2012; World Wide Web Foundation, 2015). The gap in access and use is more significant in rural areas with dispersed population and can highly affect the ability of community members to take part in public participation processes. As an example, public participation in the process of formulating a national strategy for wildlife conservation and management in 2017 was only possible by sending a written memorandum via email or expressing oral or written opinions in the 'structured regional meetings'. Sending a written memorandum via email, among other things, requires a certain level of literacy, computer skills and internet access, provided that community members were aware of this consultation in the first place. Moreover, structured meetings at the regional level are normally held at the center of each county which also limits the chance of participation for community members who live in remote areas.

6. Discussion

This section summarizes the most important findings of this study and demonstrates that aside from the fact that the two CBM initiatives in this study have different thematic foci, they are operating in fundamentally different social, political, institutional and technological settings.

Unlike the Kenyan case study, the Dutch case study can be characterized as an institutional environment that currently has little or no envisioned role for public participation, and a social context with high trust in, and expectations from, the government. This issue was also highlighted in previous research by Wehn et al. (2015), where they concluded that flood risk management in the Netherlands is considered by the public and by authorities as an issue that requires technical expertise and needs to be handled by the authorities and not by citizens. This has resulted in an evident awareness gap and reluctance of citizens to contribute to tasks related to water management (OECD, 2014; Wehn et al., 2018). Municipalities and Regional Water Authorities are the main stakeholders involved in managing pluvial flooding and the role of community members is mostly limited to electing representatives in the municipality and Regional Water Authority elections. Nevertheless, municipalities and Regional Water Authorities have occasional conflict of priorities and interests, especially when it comes to spatial planning. The fact that the upcoming 'Environmental Planning Act' combines the 'Water Act' and the 'Spatial Planning Act' into one coherent piece of legislation shows that the Dutch government is aware of this problem and is taking a step towards resolving it. Although the overall level of availability and access to data was not identified as a prominent issue in the Dutch case, the Regional Water Authorities collect and store most of the data about pluvial flooding and data sharing with other stakeholders often happens on demand. Moreover, the Netherlands ranks high in terms of access to ICTs and a large proportion of the Dutch population is actively using ICTs and has the required skills to do so.

The results of our study elicited several contextual factors in the Kenyan case study. Social, institutional, political and technological circumstances that, if not understood well, are known to act as 'major barriers' for establishing CBM initiatives in East Africa (Pocock et al., 2018) and beyond (Haklay, 2015; Tseng & Penning-Rowsell, 2012). The Kenyan case study can be characterized as a relatively new political and

institutional climate, with contested power over decisions regarding biodiversity conservation and livelihood management. Despite the constitutional authority and mandates of the Narok County Government for managing its resources, other powerful stakeholders at different levels (e.g. the national government organizations and the local chiefs) have a lot of interest in, and influence on, decision making processes. Similar to the Dutch case, the decision making system in this case study is mostly top-down; however, community members are more concerned and conscious about environmental issues, and have a much lower level of trust in authorities. Despite much emphasis on public participation in the Constitution of Kenya and several pieces of legislation, local citizens who are not well-connected or cannot take part in participatory meetings are essentially not being heard. In terms of different forms of access to ICTs, Narok County performs below the national level; a problem that among other things is amplified by the national strategy for spatial development of ICTs and the dispersed distribution of the Narok County population. Furthermore, it is evident that there is a lot of sensitivity about publishing certain data and information in this case study. As an example, publishing live data about the location of wildlife could raise concerns about potential misuse of such data by poachers for locating certain species.

7. Conclusions

In this paper, we used the CPI Framework, a framework for context analysis, processes evaluation and impact assessment of CBM initiatives, which was developed by Gharesifard et al. (2019) to systematically analyze the contextual settings during the initiation of two CBM initiatives, in the Netherlands and Kenya, within the context of the Ground Truth 2.0 project. A contextual analysis was carried out to depict the existing power dynamics, participation paradigm and technological settings in these two case studies. Based on the findings of this research, a number of realistic expectations of what the Netherlands CBM (Grip op Water Altena) and the Kenyan CBM (Maasai Mara Citizen Observatory) can achieve are identified and presented in this section. In addition, a number of recommendations are provided that can be beneficial for the future development and functioning of these two CBM initiatives. These insights will help making informed decisions while designing and implementing CBM initiatives and provide valuable inputs for their future up-scaling. Without a sound understanding of the contextual realities in which these CBM initiatives will operate, assumptions need to be made in order to understand what can be expected from each CBM initiative. Moreover, there will be no baseline for measuring outcomes and impacts of the initiatives. Therefore, it is highly important that such a baseline analysis be conducted at a very early stage and preferably before the establishment of a CBM initiative. We hypothesize that early assessment of the contextual setting in which a CBM initiative will operate and with which it will interact can help establish more sustainable initiatives; a hypothesis that requires further examination and verification.

The Dutch case study presents a case of a well-established top-down water management system. A system in which public participation is not embedded in the institutions and the general public has a low level of awareness of, little interest in, and no direct authority for managing pluvial flooding and essentially expects the authorities to manage this issue. One of the roles of the Grip op Water Altena is therefore to raise awareness and build a better understanding of the issue of pluvial flooding among Dutch citizens. In addition, this CBM can help create a more interactive pattern of communication and information sharing among water managers, spatial planners and citizens; a functionality that can help achieve the desired coordination goal of the upcoming 'Environmental Planning Act'. Although municipalities and the Rivierland Regional Water Authority are the main stakeholders involved in managing pluvial flooding in this case study, involving other stakeholders at higher levels of policy making (e.g. Rijkswaterstaat and the province of Noord Brabant), as well as the Rioned Foundation and

ZLTO, is highly recommended for the long term sustainability of this CBM. Finally, although the 'access to technology' gap is almost closed in the Netherlands, further technological development of this initiative should consider the requirements of interested citizens with limited digital skills.

The Maasai Mara Citizen Observatory has the potential to help create a much needed platform that facilitates information collection and sharing among stakeholders in the Mara region. To do so, it can leverage the existing interest within the county government, national government and private sector for improving the availability of spatial and temporal data about biodiversity and livelihoods, as well as public interest in accessing information that affects their livelihoods. Moreover, given the institutional provisions for public participation, this CBM has the potential to facilitate the process of public participation in decision making and also communication among different actors. Moreover, the identified gap in access to ICTs in this case study highlights the need for developing technologies that are open source and can be used offline. The Open Data Kit (ODK) is an example of such technological solutions; a suite of tools that allows data collection and submission to an online server without internet connection or mobile carrier service at the time of data collection. Nevertheless, because of the low average level of digital skills among the users, it is recommended that a user manual is developed and training sessions are organized to further familiarize end-users with functionalities of the App and the CBM Web platform. It is also recommended that alternative possibilities for inclusion of those community members who do not use smart phones are further explored, e.g. the possibility of dissemination of selected information via local radio. The existing concerns about sharing sensitive data may result in developing specific data collection or sharing policies that restrict access to live/raw data for the end-users. Transparency about such a decision is key for building trust among users, an issue that is important for the long-term success of a CBM. Therefore it is recommended that such decisions are clearly communicated with the stakeholders. Moreover, based on the results of this research, for scaling up the CBM and recruiting more members, it is highly recommended to take the local chiefs and churches onboard, as these are among the most influential stakeholders and can help attract a large number of community members.

Future research should trace the changes in the baseline contextual settings of the Grip op Water Altena and Maasai Mara Citizen Observatory, both during and after the lifetime of the Ground Truth 2.0 project. Moreover, application of the comprehensive framework proposed by Gharesifard et al. (2019) to fully functional CBMs will complement the results of this research by capturing the intrinsic characteristics of the two initiatives.

While having demonstrated the advantages of conducting an in-depth baseline analysis at the initiation of a CBM initiative, we also acknowledge that undertaking such a thorough contextual analysis initiatives requires substantial time, resources and expertise, which may not always be available for all CBM initiatives. Nevertheless, a less comprehensive context mapping that is tailored to available resources and based on a desk research of secondary data sources can still yield valuable insights and is therefore highly recommended.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2019.124144>.

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